

Cloudflare Radar 2024

Year in Review

A data-driven analysis of global Internet traffic, adoption trends, connectivity disruptions, and security threats observed through Cloudflare's network in 2024.

17% Global Traffic Growth and Starlink's 3.3x Surge

Global Internet Traffic Growth

Grew by 17% in 2024, driven by increasing reliance on communication, commerce, and entertainment.

SpaceX Starlink's Rapid Rise

Traffic saw dramatic 3.3x growth during 2024, bringing satellite Internet connectivity to previously underserved locations.



Google Leads Top 10 Services as Generative AI Rises

Overall Rankings

Google maintained #1 position for the second consecutive year.

Facebook, Apple, TikTok, and AWS round out the top 5.

Emerging Trends

Generative AI continued its rapid rise throughout 2024.

The Metaverse receded into the background.

Rank	Service
#1	Google
#2	Facebook
#3	Apple
#4	TikTok
#5	AWS
#6	Microsoft
#7	Instagram
#8	YouTube
#9	Amazon
#10	WhatsApp

Post-Quantum Encryption Jumps from 2% to 13% of TLS

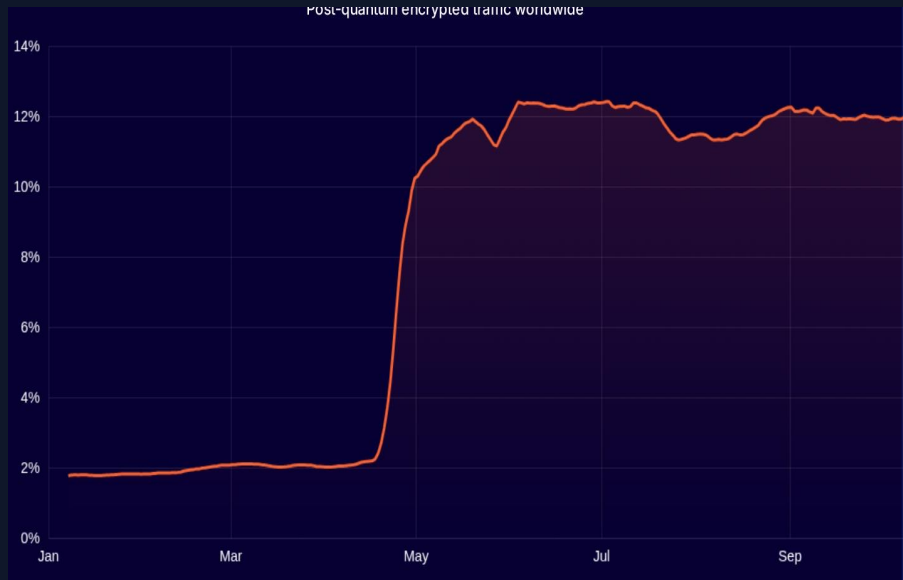
1.3

Post-Quantum Surge

Adoption jumped sharply following Google Chrome 124 enabling post-quantum key agreement by default on April 17.

Other Tech Adoption Shifts

HTTP/3 reached 21% adoption, providing faster connections and encryption by default. Go became the top API client language with approx 12% of automated API requests.



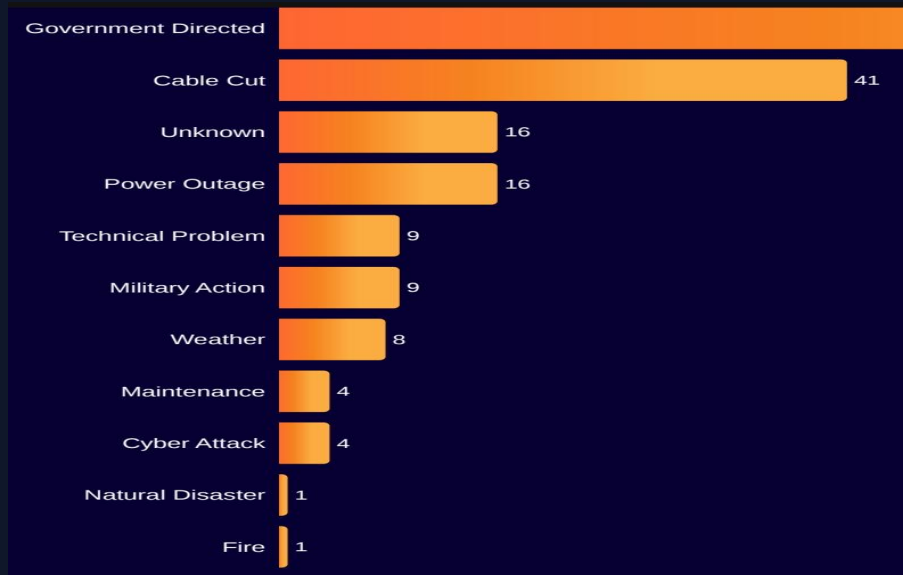
225 Major Disruptions: Government Shutdowns Lead Outage Causes

Global Connectivity Fragility

Cloudflare observed 225 major Internet disruptions globally in 2024.

Leading Causes

- Government Directed (116+ incidents)
- Cable Cut (41 incidents)
- Power Outage (16 incidents)



6.5% of Traffic Mitigated: Persistent Threats and Bot Concentration

6.5%

of all global traffic was mitigated

Gambling/Games

Most attacked industry category

4.3%

of processed emails were malicious

Bot Traffic Concentration

68% of global bot traffic originated from just the top 10 countries.

Persistent Vulnerabilities

The Log4j vulnerability (CVE-2021-44228) continues to be actively targeted more than two years after disclosure.

Five Takeaways Shaping the 2025 Internet Outlook

Traffic Growth Trajectory

Global Internet traffic grew 17% year-over-year, with satellite connectivity surging to serve new regions.

Post-Quantum Encryption Momentum

Adoption leapt to 13% of TLS 1.3 traffic following browser default enablement.

Connectivity Fragility from Government Actions

Government-directed shutdowns remained the leading cause of major Internet disruptions.

Persistent Security Threats

Threats remain pervasive across web traffic (6.5% mitigated) and email vectors (4.3% malicious).

Evolving Developer Ecosystem

Go has surpassed other languages to become the most popular API client language (roughly 12%).